

Trajectory Optimization for 6G-UAV Based on Deep Reinforcement Learning

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Abstract—With more and more distributed deployment of Internet of Things (IoTs) in 5/6G networks, the unmanned aerial vehicles (UAVs) have attracted much interest for mobile edge computing service outside the coverage of terrestrial cellular systems. However, due to the interference and coverage hole issues of UAV-assisted aerial 3D communications, how to design a suitable UAV trajectory is crucial for the performance of future 6G-UAV. In order to achieve a balance between the communication quality and mission accomplishment, we define an environmental model with random interference and formulate a trajectory optimization problem to improve the communication coverage performance in high mobility conditions. By setting an interference point in the highly dynamic networks, we propose a novel trajectory optimization algorithm based on deep reinforcement learning (DRL) to minimize the overall task execution delay and expected outage duration. The received data information of UAVs is used for double deep q -network (DDQN) training while the prioritized empirical replay architecture in the prioritized replay deep q network (DQN) is utilized to select the best UAV action, avoid the UAV flying into the random interference points, and reduce the random interference. Simulation results demonstrate the superiority of the proposed algorithm.

Index Terms—UAV, deep reinforcement learning (DRL), trajectory optimization.

I. INTRODUCTION

With the arrival of 5G/6G era, the unmanned aerial vehicle (UAV) communication becomes a hot research field to assist the wireless mobile communication as air base station or relay. Its main applications lie in two aspects: UAV-assisted wireless communication and cellular-connected UAV [1], [2]. UAV-assisted wireless communication can take as a flying base station or mobile relay to enhance the communication services for the ground users in the event of high traffic demand or damage of ground base stations. Cellular-connected UAV takes the UAV as a new air user to access the traditional cellular network and enable a variety of applications, such as post-disaster rescue, industrial monitoring, and agricultural detection [3]. In this paper, we mainly focus on the second application scenario. Yet, the mobile service coverage of UAV can be effected greatly by the trajectory design [1], [4].

Significant existing research works factoring in the trajectory design challenge have been proposed for enabling the UAV communications. Considering the cooperative sensing and transmission for UAV trajectories, a distributed sense-and-send protocol was proposed by selecting from a discrete set of tasks and a continuous set of locations for sensing and transmission [4]. In [5], a random deep reinforcement learning (DRL) strategy was proposed for the flight navigation of multiple UAVs. In [6], DRL was used to allow UAVs to learn the environment

dynamics and make good trajectory decisions for post-disaster rescue efforts. In [7], a path optimization problem for multiple UAVs was managed by using block coordinate descent (BCD), successive convex approximation (SCA), and S-program, while the high computational difficulty and low accuracy weren't handled well. Furthermore, a UAV coverage navigation algorithm based on communication coverage of ground base stations was proposed where the multi-step DRL was utilized [8]. However, most of the above existing technologies did not jointly consider the multiple access interference, communication quality, anti-interference, and flight efficiency to achieve a balance. The multiple access interference from various terminals will seriously affect the communication quality of users.

From the above factors, we investigate a novel DRL based anti-interference UAV trajectory optimization algorithm by Markov decision process (MDP). Under the condition that the UAVs aim to fly from a random initial point to a given mission end with the minimum time and acceptable service quality, we also study the multiple access interference from not only high radiation strong signals but also ground buildings in line-of-sight links to the UAV communications by setting several random interference points. We use the priority replay DQN framework to analytically select the action-state gain value to avoid flying into the disturbed area as much as possible and reduce the interference impact from the random jamming points. The two main contributions in such a design are (i) except for the interference from ground buildings in the line-of-sight links, we also study the interference from high radiation or other strong signals and thus set a number of random interference points to simulate the practical situations, and (ii) in order to reduce the interference from the random jamming points, we use the priority replay DQN framework to analytically select the action-state value and reduce the impact from the random jamming points as much as possible.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

Consider an airspace with a three-dimensional cube, denoted by R_{LH} , where L and H denote the lower and upper bounds of the three-dimensional airspace coordinates, respectively. There are several terrestrial cellular base stations and several random jamming points in the flight area of UAVs [7], [8], which will impact the airborne connections and make it closer to the actual complex environment. We assume that the UAVs fly from a random initial position, q_C , to the mission receiver point, q_Z , for aerial detection or aerial surveillance [9]. The random interference points are unknown at the beginning. But every flight of UAVs from the beginning to the end, i.e., one episode, will accumulate the corresponding experience which can be used for the next iterative update of the flight path. For a better reading of this paper, the important symbols are listed in Table I.

In order to maintain a certain quality of service and reach the mission destination as quickly as possible, the optimal flight trajectory of UAV should be related with the location of cellular base stations and random interference points [10], i.e., as close as possible to the cellular base stations and as far as possible from the random interference points. Thus, we set the scenario as follows.

$$q(0) = q_C, q(T) = q_Z, \quad (1)$$

$$q(t) \in R_{LH}, \forall t \in (0, T), \quad (2)$$

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TABLE I
IMPORTANT SYMBOLS

Symbol	Description	Symbol	Description
R_{LH}	Simplified 3 dimensional cube	q_C	Random initial position
q_Z	Mission receiver point	$q(0)$	Initial UAV position
$q(T)$	Final position of UAV	$q(t)$	UAV position at t with norm of 1
$p_r(t)$	Received power of UAV at t	$\bar{P}_b$	Transmit power of cellular cell b
$\beta_b(\cdot)$	Large scale channel gain	$G_b(\cdot)$	BS antenna gain
$\bar{h}_b(t)$	Small scale fading variable	T_{off}	Communication interruption time
P_{off}	Communication interruption state	$\hat{v}(s, \omega)$	State value function
$\hat{q}(s, a, \omega)$	Action value function	γ	Learning rate
P_{out}	Out-of-bound penalty	R_{arr}	Arrival bonus
R_{jam}	Interference penalty	$P_p(SIR(t) < \xi)$	Outage probability
Z_i	Random interference point	T_{fly}	Total flight duration
ω_j	Priority weight of the j th sample	$Q(q_m, V; \theta)$	Training value
$Q(q_m, V)$	True value	-	-

where T denotes the mission execution time and $q(t)$ denotes the UAV trajectory at time t .

B. Channel Model

Based on the free-space propagation model in [8], [11], the received power of UAV at t , $p_r(t)$, can be expressed as

$$p_r(t) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}, \quad (3)$$

where P_t is the transmit power, G_t is the transmit antenna gain, G_r is the received antenna gain, d is the distance between the transmitter and receiver, L is the propagation independent channel loss factor, and λ is the wavelength.

From the above, the received power of UAV is related with the base station transmit power, transmit antenna gain, receive antenna gain, large scale fading, small scale fading, and interference, which can be effected by the UAV location. Therefore, the above expression can be rewritten as

$$p_r(t) = \bar{P}_b |h_b(t)|^2 = \bar{P}_b \beta_b(q(t)) G_b(q(t)) \bar{h}_b(t), b = 1, \dots, B \quad (4)$$

where $\bar{P}_b$ denotes the transmit power of cellular cell b . B denotes the maximum number of cellular cells. $\beta_b(\cdot)$ and $G_b(\cdot)$ denote the large scale channel gain and BS antenna gain, respectively. $\bar{h}_b(t)$ is a random variable related with the small scale fading.

If the received signal-to-noise ratio at destination is below a certain threshold, the UAV will enter into a disconnected state, P_{off} . So, the disconnection time T_{off} can be expressed as

$$\begin{aligned} T_{off}(\{q(t), b(t)\}) &= \int_0^T P_{off}(q(t), b(t)) dt \\ &\triangleq \int_0^T P_p(SIR(t) < \xi) dt, \end{aligned} \quad (5)$$

where $b(t)$ denotes the cellular cell b connected with the UAV at t and $P_p(SIR(t) < \xi)$ denotes the outage probability.

In order to analyse the interference effect, we set Z_i as the random interference point. For simplicity, the interference strength is set to the maximum value firstly, i.e.,

$$P_{cp} = \begin{cases} \bar{P}_{off}, & \text{if } q(t) \notin Z_i \\ 0, & \text{if } q(t) \in Z_i \end{cases} \quad (6)$$

where P_{cp} denotes the coverage probability and $\bar{P}_{off}$ represents the complementary outage probability.

C. Problem Formulation

Based on the previous analysis, the trajectory optimization problem of UAV can be formulated as follows.

$$P(1) : \min_{T, \{q(t), b(t)\}} T_{fly} + \alpha T_{off} \quad (7)$$

$$s.t. \quad \|\dot{q}(t)\| \leq V_{\max}, \forall t \in (0, T), \quad (7a)$$

$$q(0) = q_C, q(T) = q_Z, \quad (7b)$$

$$q(t) \in R_{LH}, \forall t \in (0, T), \quad (7c)$$

$$b(t) \in \{1, \dots, B\}, \quad (7d)$$

where α is the weight coefficient. (7a) aims to restrict the UAV maximum flight speed. (7b) aims to define the flight starting and ending points of UAV. (7c) aims to restrict the UAV flight area as requested. (7d) represents the cellular cell sequence.

If the flight speed of UAV achieves the maximum value, the optimal solution of P(1) can be obtained correspondingly, i.e., $\dot{q}(t)$ in (7c) can be expressed as $\dot{q}(t) = V_{\max} \vec{v}(t)$, where $V_{\max}$ is the maximum velocity vector and $\vec{v}(t)$ denotes the UAV flight direction. When $\|\vec{v}(t)\| = 1$, P(1) can be rewritten as P(2) as follows.

$$P(2) : \min_{T, \{\dot{q}(t), \vec{v}(t), b(t)\}} T_{fly} + \alpha T_{off} \quad (8)$$

$$s.t. \quad \dot{q}(t) = V_{\max} \vec{v}(t), \forall t \in (0, T), \quad (8a)$$

$$\|\vec{v}(t)\| = 1, \forall t \in (0, T), \quad (8b)$$

$$(7b) - (7d). \quad (8b)$$

Since the above trajectory optimization problem is time-space continuous and the amount of involved data is large, it is difficult to obtain the optimal solution by the traditional algorithms, such as convex optimization, graph theory, and etc. They require huge time and computational cost, and the problem often is non-convex. In this paper, we will find the optimal trajectory optimization solution with the assistance of DRL which includes the popular dueling DDQN and priority replay DQN algorithms [9], [12].

III. PROPOSED UAV TRAJECTORY OPTIMIZATION ALGORITHM

A. DRL Model Design

As we all know, the finite discrete state sets in reinforcement learning process can be used to solve the small scale sequential decision problems. However, when the state is continuous or the discrete set is with a large amount of data, the traditional learning methods, such as Q-learning, are not suitable. At this time, a feasible modeling solution is

the approximate representation of value function. So, in this paper, we define a state value function, $\hat{v}$, which can be expressed with parameter, ω . Taking the state, s , as input and the strategy, π , as action decision, we obtain the value function of state, s , i.e.,

$$\hat{v}(s, \omega) \approx v_\pi(s). \quad (9)$$

Similarly, the action value function, $\hat{q}$, can be calculated with parameter, w . Taking the state, s , and action, a , as input, we can obtain the action value function as follows.

$$\hat{q}(s, a, \omega) \approx q_\pi(s, a). \quad (10)$$

Many existing approximate methods can be used for the above values, such as linear representation, Fourier transform, and so on. Among them, the artificial neural network (ANN) approach is the most effective one which calculates the input signal and generates the output through a nonlinear activation function [11]. Reinforcement learning combined with the deep ANN function approximation brings out a powerful DRL framework [12]. Thus, in our paper, we approximate $Q(s, a) \approx Q(s, a; \theta)$ as an action value function in DNN, where θ can be updated based on the classical back-propagation algorithm [12]. Then, the state parameters are trained by bootstrapping and for each given state-action-reward-next state transition (S_n, A_n, R_n, S_{n+1}) , the parameter, θ , can be updated with the minimizing loss.

We further define a new target network, $\hat{Q}(s, a; \theta^-)$, to train the parameters in DRL model, where $\hat{Q}(s, a; \theta)$ comes from Q network and θ^- is updated with low frequent. So, the loss function can be expressed as

$$Loss = (R_m + \gamma \max_{a \in A} \hat{Q}(S_{m+1}, a; \theta^-) - \hat{Q}(S_m, A_m; \theta)). \quad (11)$$

Note that other DRL algorithms can also obtain the equivalent performance, such as DDQN which uses two deep learning networks to reduce the learning bias, DQN with the prioritized experience replay [12], DDPG [13], and A3C [14].

B. UAV Trajectory Optimization Problem Solution

Similar with [11], we set the random jamming points as Z_i , where i is the number of jamming points. Considering an extreme case to verify the anti-interference performance of our proposed algorithm, we assume that the communication efficiency is reduced to 0 when the normal communication is not possible. Based on [8], we take the UAV trajectory optimization problem as a sequential MDP Markov decision issue. Thus, $P(2)$ can also be transformed into

$$P(3) : \max_{M, \{q_m, \vec{v}_m\}} -M - \alpha \sum_{m=1}^M \hat{P}_{off}(q_m) \quad (12)$$

$$s.t. \quad q_{m+1} = q_m + \vec{v}_m \xi s, \forall m, \quad (12a)$$

$$\|\vec{v}_m\| = 1, \forall m, \quad (12b)$$

$$q_0 = q_C, q_M = q_Z, \quad (12c)$$

$$q_m \in R_{LH}, \forall m, \quad (12d)$$

where $\xi s = V_{max} \xi t$. The total task time T is divided into M slots ξt , i.e., $T = M \xi t$, where ξt needs to be small enough so that the position of UAV approximates to a constant in each time slot. The natural mapping of $P(3)$ to MDP $\langle S, A, P, R \rangle$ can be done as follows.

S : The state space corresponds to the UAV flight space, i.e., the state relates with all the possible positions of UAV, $S = \{q, q \in R_{LH}\}$. Besides, we also define that the out of bounds indicates UAV out of the flight space. The interference indicates UAV within the random interference area and the arrival indicates UAV in the end position.

A : The action selection space can be expressed as $A = \{\vec{v}, \|\vec{v}\| = 1\}$, which represents the UAV flight direction. Both the action and state sets are continuous.

P : The state transfer probability, $P(s'|a, s)$, denotes the probability of transferring to state s' after taking action a in state s , specifically, based on (12a).

R : The reward is defined as a function of the received signal strength and the different reward functions are designed based on the different UAV states as follows.

$$r = \begin{cases} P_{out}, & \text{if } q_m < q_L \text{ or } q_m > q_H, \\ R_{arr}, & \text{if } q_m = q_Z, \\ P_{jam}, & \text{if } q_m \in Z_i. \end{cases} \quad (13)$$

where the three possible rewards are: 1) when the UAV flies outside the flight space range, it will receive an out-of-bounds penalty, where the weight of the transboundary penalty is α ; 2) when the UAV reaches the end point, it will receive an arrival bonus; 3) when the drone flies into the area affected by a random interference point, it will receive a penalty.

C. Prioritized Replay DQN Algorithm With Dueling DDQN for UAV Trajectory Optimization

We discretize the action space A into I with $A = \{v_1, v_2, \dots, v_I\}$, where $I = 4$ and $v_I = \{N, S, E, W\}$ refer to four flight directions: north, south, east, and west, respectively. In order to solve the above trajectory optimization problem, we propose a prioritized replay DQN algorithm with dueling DDQN learning based on DRL, as shown in Algorithm 1. Firstly, we formulate a dual Q -network in which one is for evaluation and the other is for target network. During the learning training process, the samples are selected by the priority replay DQN and the absolute TD error is calculated for each sample which is used for setting the priority, i.e., the higher sample value, the easier to get trained. Here, TD error refers to the difference between the true and estimated values in system model. To avoid invalid sample training to affect the algorithm convergence speed, SUMTREE structure store memory is adopted here.

Compared to the traditional DQN algorithm, the proposed algorithm uses duel output branches, the state scalar value, V , and the dominance vector value, A , and constructs two action value functions, i.e., one for estimating the action and the other for estimating the action value. It uses dueling DQN to estimate the payoff evaluation network and determine the action and the target network to determine the action value. The training coefficient of the dueling DQN, denoted as θ , is trained so that the training value $Q(q_m, V; \theta)$ performs better compared to the true value $Q(q_m, V)$. Hence, the loss function of the proposed algorithm can be calculated as follows.

$$Loss_1 = (R_m + \gamma_m \hat{Q}(s_m, \vec{v}^*; \theta^-) - \hat{Q}(s_m, \vec{v}_m; \theta))^2, \quad (14)$$

where γ is the learning rate and

$$\vec{v}^* = \arg \max_{\vec{v}' \in \hat{A}} \hat{Q}(s_m, \vec{v}'; \theta). \quad (15)$$

Based on the priority replay DQN learning with sample priority, a new loss function is designed as

$$Loss_2 = \frac{1}{k} \sum_j^k \omega_j [R_m + \gamma_m \hat{Q}(s_m, \vec{v}'; \theta^-) - \hat{Q}(s_m, \vec{v}_m; \theta)]^2, \quad (16)$$

where ω_j is the priority weight of the j th sample which is obtained by normalizing the TD error $|\delta(t)|$.

Algorithm 1: Proposed Trajectory Optimization Algorithm.

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1: Initialize:  $M_{epi}, M_{step}, D_{tol}, k, K, N, H = \phi, \sigma, \tau, \Delta = 0,$ 
    $p_1 = 1, \varepsilon_0, \mu, \varepsilon \leftarrow \varepsilon_0, R_{arr}, P_{out}, P_{jam}, \alpha, \theta, \text{ and } \theta^- = \theta$ 
2: for  $m_{epi} = 1, 2, \dots, M_{epi}$  do
3:   form  $(S_m, R_m, \gamma_m), (q_{m-1}, v_{m-1}, R_m, \gamma_m, q_m)$ 
   with  $p_m = \max_{i < m} p_i, q_C \in S,$ 
4:   set the time step  $m \leftarrow 0$  and  $q_0 \leftarrow q_C$ 
5:   repeat
6:     if  $m \equiv 0 \bmod K$ 
7:       for  $j = 1, 2, \dots, k$  do
8:         choose  $v_m$  from  $A$ , i.e.,  $v_m = v(I^*) \quad I^* =$ 
            $\begin{cases} randi(I), with prob. \varepsilon \\ \arg \max_{i=1, \dots, I} \hat{Q}(q_m, \vec{v}^{(i)}; \theta), with prob. 1 - \varepsilon \end{cases}$ 
9:         execute  $v_m$ , observe  $q_{m+1}$ , and obtain
            $P_{off}(q_{m+1})$ , with  $R_m = -1 - \alpha P_{off}(q_{m+1})$ 
10:        sample  $(q_j, v_j, R_j, \gamma_j)$  from  $H$ 
11:        sample transition  $j - p(j) = p_j^\sigma / \sum_i p_i^\sigma$ 
           compute  $\omega_j = (N \cdot P(j))^{-\tau} / \max_i \omega_i$ 
           compute TD-error  $\delta_j = R_j + \gamma_j \hat{Q}(s_j, \vec{v}^*; \theta^-) -$ 
            $\hat{Q}(s_{j-1}, \vec{v}_{j-1}; \theta)$ 
           update transition priority  $p_j \leftarrow |\delta_j|$ 
12:       end for
13:       perform a gradient descent step on  $\theta$ 
14:     end if
15:     update  $m \leftarrow m + 1, \varepsilon \leftarrow \varepsilon \mu$ 
16:   until  $\|q_m - q_Z\| \leq D_{tol}$  or  $q_m \notin S$ , or  $m = M_{step}$ 
17:   copy weights into target network  $\theta^- \leftarrow \theta$ 
18: end

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IV. SIMULATION RESULTS AND DISCUSSION

In this section, we evaluate the effectiveness of our proposed algorithm with numerical results. As comparison, we also evaluate the performance of direct reinforcement learning algorithm, duel DDQN algorithm, and randomized algorithm with randomly selected actions. The implementation of simulation setup based on DRL framework is detailed in [9] and the SumTree binomial tree is used as the empirical playback pool with original sample priority, $p_1 = 1$. In addition, the UAV executes its mission in a three-dimensional space of (1000, 1000, 100) and its starting point is random while the end point is located at (1400, 1600, 100). The statistical city building model and wireless communication antenna array model are similar with [8], where there are 7 ground base stations and random interference points. The building heights are not exceeding 90 meters, as shown in Fig. 1.

Fig. 2 demonstrates the signal coverage probability map which is the ratio of complementary to interruption probability. Fig. 2(a) and (b) show the effect of different numbers of random interference points on the signal coverage probability in the UAV mission space, respectively. It can be seen that the signal coverage is related with the distribution location of base stations, different heights of buildings, and locations where the random interference points appear. Our proposed algorithm aims to avoid the UAV passing through the areas with weak signal coverage as much as possible and reach the preset end point as soon as possible.

Fig. 3 shows the dynamic reward comparison of different algorithms under the same random interference points. It can be seen that the performance fluctuation of proposed method is better than the direct reinforcement learning algorithm while similar with the dueling DDQN algorithm. The payoff reward is better than the others after 5000 sets. This is because the motion state of UAV is continuous and the empirical replay priority structure of prioritised replay DQN facilitates

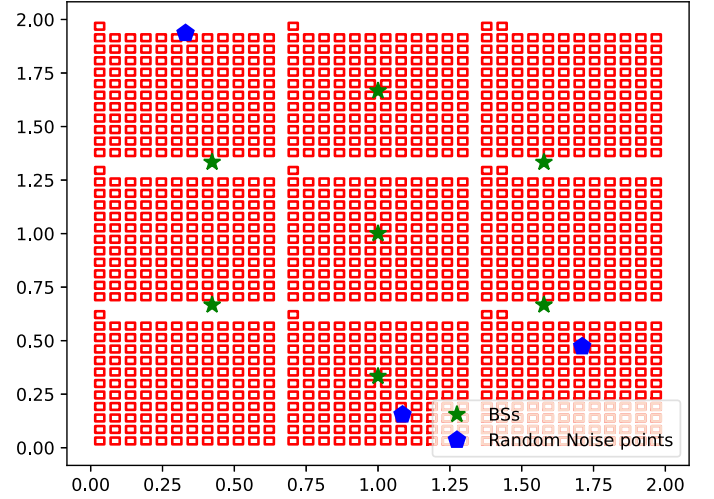


Fig. 1. Two dimensional schematic of UAV mission space.

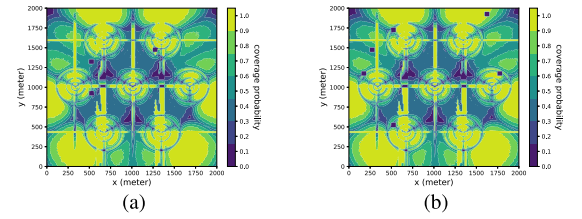
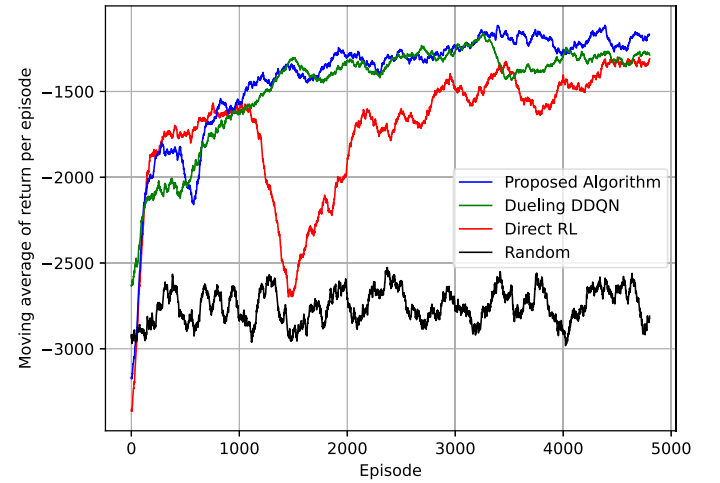
Fig. 2. Signal coverage with different random jamming points. (a) Jamming point $Z_i = 3$. (b) Jamming point $Z_i = 6$.

Fig. 3. Average return comparison of different algorithms.

the algorithm convergence, resulting in more efficiency of proposed algorithm.

Moreover, by defining the endpoint arrival rate as the ratio of total transmission time slots to distinct episodes, we compare our proposed algorithm with the benchmark algorithms in Fig. 4, i.e., dueling DDQN, direct reinforcement learning, and random action selection. It can be seen that except for the random algorithm, the others all increase the endpoint arrival rates when the total episode number increases, which reach the maximum after 5000 episodes. Our proposed algorithm has a higher value than the direct reinforcement learning in 1000 - 5000

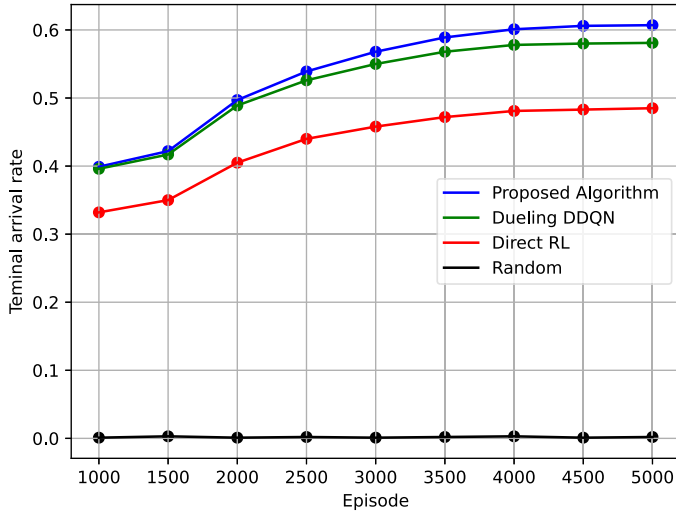


Fig. 4. Comparison with different end point arrival rates.

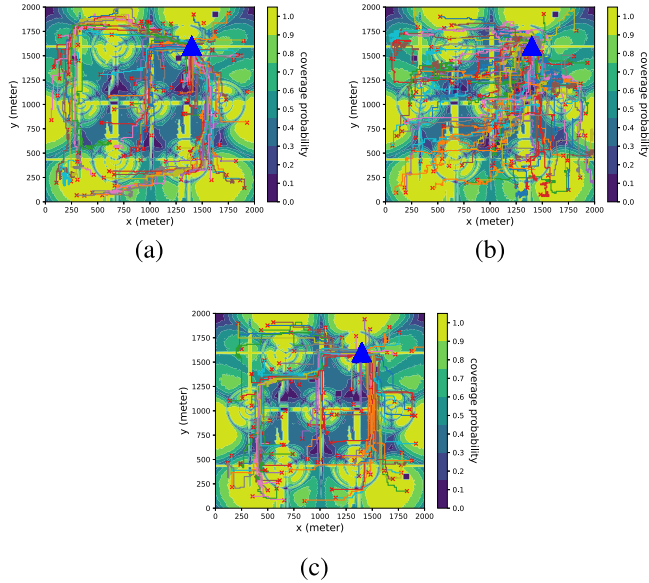


Fig. 5. Signal coverage of different number of random jamming points. (a) $Z_i = 3$, $M_{\text{epi}} = 1000$, $P_{\text{jam}} = 500$, $\alpha = 30$. (b) $Z_i = 3$, $M_{\text{epi}} = 3000$, $P_{\text{jam}} = 100$, $\alpha = 5$. (c) $Z_i = 6$, $M_{\text{epi}} = 5000$, $P_{\text{jam}} = 500$, $\alpha = 30$.

episodes, and a slightly higher than the dueling DDQN in 2000–5000 episodes.

Fig. 5 plots the UAV flight trajectory for the last 100 episodes of our proposed algorithm. Both Fig. 5(a) and (b) show that the smaller the interference penalty, P_{jam} , and out-of-bounds penalty factor, α , the easier the UAV flying to the preset endpoint. This is because that the flying process within the random interference points and signal-weak areas has less effect on its penalty and the UAV tends to reach the preset endpoint faster. The comparison of Fig. 5(a) and (c) shows that the UAV is easier to avoid the random interference points and signal weak areas when the total episode of training iterations reaches about 5000. The flight path tends to consistent when the total episode is increasing.

V. CONCLUSION

In this paper, we have developed a novel trajectory optimization algorithm for future 6G-UAV with cellular connections based on DRL and random interference point setting. By formulating MDP model and sample priority DQN/dueling DDQN, we have designed the output of UAV flight maneuvers. Of course, the proposed solution still has some limitations in terms of implementation and complexity, such as without considering (i) the overall UAV deployment, and (ii) the onboard energy loss of UAV. In the future, we will continue to look into them.

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